SINDH · INTERMEDIATE / CLASS XI · COMPUTER SCIENCE

Lecture-wise Study Guide

Twenty-five college sittings with outcomes, board work, exit tickets, and homework. MathJax notation and SVG diagrams for Digital Logic, SDLC, Networks, Algorithms, Python, Databases, IoT & AI, and Digital Literacy.

One lecture · One artefact · One recitation

How to use this lecture edition

Written for Intermediate / HSC Class XI students following the Sindh Textbook Board Computer Science syllabus. Each sitting is one college lecture: outcomes, taught content, a board example, an exit ticket, and a 15-minute homework. Formulae are MathJax; diagrams are SVG. This is a companion, not a reprint — keep your textbook for official figures and LogiSim screenshots.

THE FIVE-PASS METHOD

1. **Read for the map** — the outcomes box, two minutes.
2. **Read for the mechanism** — cover a worked line and try it first.
3. **Exit ticket** is a mini exam; write on paper.
4. **Recite** the lecture's one list (gates, CAPP, OSI, keys…).
5. **Homework tonight, review next week.**

Twenty-five sittings · eight college weeks

Week	Lectures	Must-finish artefact
1	L01–L03	LogiSim AND + the seven-gate table from memory
2	L04–L06	One SOP table and the SDLC list closed-book
3	L07–L09	Waterfall vs Agile from memory; CT strategies
4	L10–L12	Two sort traces and one binary-search diary
5	L13–L15	Calculator + even/odd running in VS Code
6	L16–L18	ATM or guess game; four keys recited
7	L19–L21	One ER + one grouped query on paper
8	L22–L25	Irrigation paragraph + 40-mark rehearsal

Lecture register

No.	Title	Syllabus
L01	Stairs and ramps — analog versus digital	Unit 1
L02	Boolean algebra — arithmetic with truth	Unit 1
L03	The seven logic gates	Unit 1
L04	Truth tables, SOP, and logic diagrams	Unit 1
L05	Karnaugh maps — grouping the ones	Unit 1
L06	Software Development Life Cycle	Unit 1
L07	Waterfall versus Agile — Al-Noor cases	Unit 1
L08	How machines converse — OSI and TCP/IP	Unit 1
L09	Think so a machine can follow you	Unit 2
L10	Bubble Sort — neighbours swap	Unit 2
L11	Selection Sort — park the minimum	Unit 2
L12	Linear search and binary search	Unit 2
L13	Python, the IDE, and input() is a string	Unit 3
L14	Operators — including the bitwise family	Unit 3
L15	Selection — if, elif, else	Unit 3
L16	Repetition — for, while, break, continue	Unit 3
L17	Libraries, errors, and the ATM studio	Unit 3
L18	Data, DBMS, and the four keys	Unit 4
L19	ER models and the three cardinalities	Unit 4
L20	Referential integrity and Access objects	Unit 4
L21	Queries, totals, and charts	Unit 4
L22	IoT collects, analytics explains, AI decides	Unit 5
L23	Sources, impacts, and assistive technology	Unit 5
L24	Digital literacy — find, judge, collect, show	Unit 6
L25	Revision sitting — cards, alarm, 40-mark paper	Revision

> **WHAT “DONE” LOOKS LIKE**
> You can teach the lecture’s core idea to a classmate in three minutes. You have hand-traced the board example. You scored the exit ticket closed-book and corrected every miss in writing.

L01

UNIT 1 · COMPUTER SYSTEMS · 45 MIN · INTERMEDIATE / CLASS XI

Stairs and ramps — analog versus digital

TODAY YOU WILL BE ABLE TO

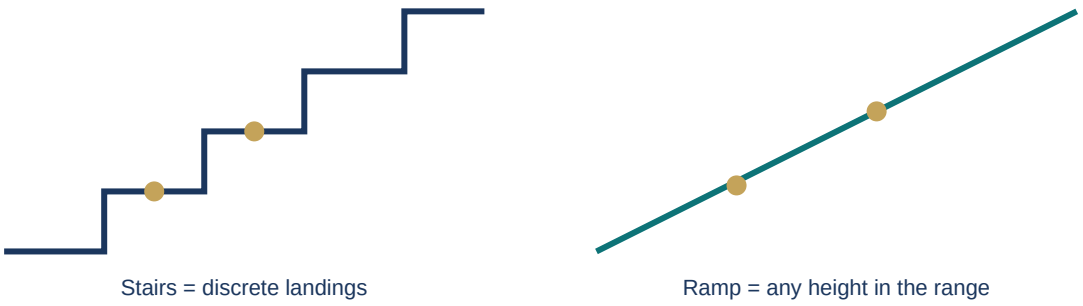
- Tell a discrete quantity from a continuous one.
- Read a sine-like analog wave and a two-level digital wave.
- Explain why a noisy digital 1 is rewritten as a clean 1.

OPENING PICTURE

A classroom has 31 students, never 31.4. Temperature can be 31.4 °C. That is the whole lecture in one pair of examples.

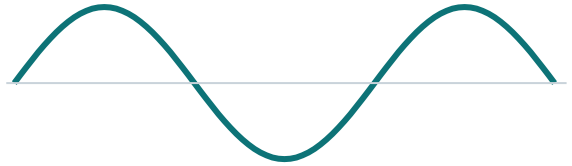
Discrete landings, continuous ramps

A **discrete** quantity takes separate countable values. You can point to each one. A **continuous** quantity can sit anywhere in a range: height, time, speed, voltage on a microphone wire.

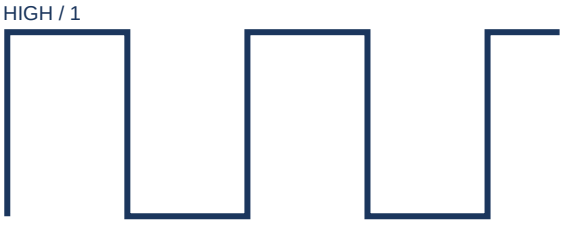


Board sketch — stairs (digital / discrete) versus a ramp (analog / continuous).

Digital electronics copies the stairs. A wire is treated as LOW (**0**) or HIGH (**1**). Everything between is noise, not a third legal value. That is why a USB copy of a photo stays sharp: a slightly damaged 1 is still recognised as 1 and rewritten as a perfect 1.



Analog — continuous wave



Digital — square wave

Figure L1.1 — Analog varies smoothly; digital jumps between two legal levels.

Feature	Analog / continuous	Digital / discrete
Nature	Smooth; infinitely many in-between values	Separate steps; usually two levels
Waveform	Sine-like	Square
Unit	A physical quantity (voltage, °C)	Bit: 0 or 1
Noise	Adds permanently	Ignored if it stays inside a band
Copying	Quality falls each copy	Perfect copies
Examples	Voice, vinyl, mercury thermometer	USB, HDMI, a calculator

BOARD TRAP

Temperature *is* continuous. A digital sensor *stores* it as bits. The phenomenon and the representation are different questions.

EXIT TICKET (4 min, closed book)

1. How many legal levels has a digital signal? 2. Why does a slightly damaged digital 1 still survive? 3. Name one analog and one digital example from this room.

HOMEWORK (15 min)

Copy the comparison table from memory. Then write two sentences: “A school bell is digital because… A ceiling fan speed (if it turns smoothly) is analog because…”

L02

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Boolean algebra — arithmetic with truth

TODAY YOU WILL BE ABLE TO

- Evaluate AND, OR, and NOT on given bits.
- Apply precedence: parentheses → NOT → AND → OR.
- Show the three intermediate values of a compound expression.

LAST LECTURE

Digital means two legal levels, 0 and 1. Today those two levels become an algebra.

George Boole (1850s) built an algebra with only $\mathbf{FALSE} = 0$ and $\mathbf{TRUE} = 1$. Variables A, B, C hold one of those values. Three primitive operations generate every digital circuit you will meet this year.

- **AND ($\cdot$)** — $Y = A \cdot B = 1$ only if *every* input is 1. Two locks on one door.
- **OR ($+$)** — $Y = A + B = 1$ if *any* input is 1. Two doors into one room.
- **NOT (A' or $\overline{A}$)** — flips one bit. A light switch.

$$(\,) \;\rightarrow\; \mathbf{NOT}\,(\,') \;\rightarrow\; \mathbf{AND}\,(\,\cdot\,) \;\rightarrow\; \mathbf{OR}\,(+).$$

Worked evaluations — write the middle steps

A. $Y = A' + B$ with $A = 1,\ B = 0$. First $A' = 0$. Then $0 + 0 = 0$. So $Y = 0$.

B. $Y = (A + B) \cdot C$ with $A = 1,\ B = 0,\ C = 1$. Parentheses: $1 + 0 = 1$. Then $1 \cdot 1 = 1$. So $Y = 1$.

C. $Y = A \cdot B + C'$ with $A = B = C = 1$. NOT first: $C' = 0$. AND: $1 \cdot 1 = 1$. OR: $1 + 0 = 1$.

DO NOW (on paper)

Evaluate $Y = (A \cdot B)' + C$ for $A = 1, B = 1, C = 0$. Write NAND first, then the OR. Answer: NAND of $1, 1$ is 0; $0 + 0 = 0$.

COMMON MISTAKE

Treating $+$ as ordinary plus. In Boolean algebra $1 + 1 = 1$, not 2. There is no ‘2’ on a digital wire.

EXIT TICKET

1. State precedence in four words. 2. Evaluate $Y = A' + B$ for $A = 0, B = 0$. 3. Why is NOT called unary?

HOMEWORK

Make a three-column table: expression / values / Y for the three worked examples, covering the answers first.

L03

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The seven logic gates

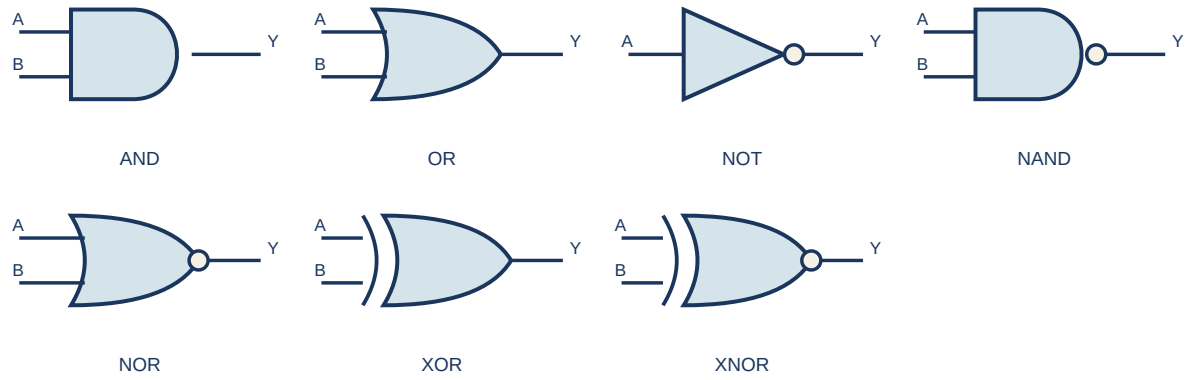
TODAY YOU WILL BE ABLE TO

- Draw AND, OR, NOT, NAND, NOR, XOR, XNOR from memory.
- Fill a 2-input truth table in binary order.
- Say why NAND and NOR are called universal.

LAST LECTURE

AND needs every 1; OR needs any 1; NOT flips. A gate is that algebra in hardware: one or more inputs, exactly one output.

With n inputs a truth table has 2^n rows. Always list inputs in binary counting order so no row is skipped: $00, 01, 10, 11$.



Derived identities you must be able to write:

$$Y_{\text{NAND}} = (A \cdot B)', \qquad Y_{\text{NOR}} = (A + B)',$$
$$Y_{\text{XOR}} = A \oplus B = A'B + AB', \qquad Y_{\text{XNOR}} = (A \oplus B)' = AB + A'B'.$$

A	B	AND	NAND	OR	NOR	XOR	XNOR
0	0	0	1	0	1	0	1
0	1	0	1	1	0	1	0
1	0	0	1	1	0	1	0
1	1	1	0	1	0	0	1

NAND and **NOR** are *universal*: any Boolean function can be built from only NAND, or only NOR. XOR is the difference detector (two-way landing light). XNOR is the sameness detector.

RECITE

XOR = eXclusively different. NAND is NOT-AND: the only 0 is the AND's only 1. NOR is NOT-OR: the only 1 is the OR's only 0.

EXIT TICKET

1. When is XOR equal to 1? 2. Write $A \oplus B$ as a sum of products. 3. Why do chip designers like NAND?

HOMEWORK + LOGISIM

From a blank page, redraw all seven symbols and the 2-input table. Then in LogiSim Evolution build a 2-input AND, poke both switches, and screenshot the four rows.

L04

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Truth tables, SOP, and logic diagrams

TODAY YOU WILL BE ABLE TO

- Build an 8-row table from a Boolean expression.
- Write a sum-of-products expression from the 1-rows.
- Draw a gate diagram that respects precedence.

LAST LECTURE

Seven gates, 2^n rows, NAND/NOR are universal. Today we travel both ways: expression → table, and table → expression.

For $Y = A \cdot B + C$ there are three variables, so eight rows. Add a helper column for $A \cdot B$, then OR with C.

A	B	C	$A \cdot B$	$Y = A \cdot B + C$
0	0	0	0	0
0	0	1	0	1
0	1	0	0	0
0	1	1	0	1
1	0	0	0	0
1	0	1	0	1
1	1	0	1	1
1	1	1	1	1

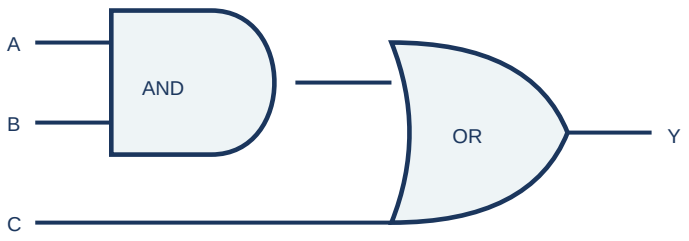


Figure L4.1 — AND is drawn first; its output and C feed the OR.

Sum of products: each row where $Y = 1$ is a **minterm**. AND every variable; complement a variable if that row's bit is 0. Then OR the minterms.

$$Y = \sum \text{minterms of the 1-rows.}$$

Example: 1-rows $AB = 01$ and 11 give $Y = A'B + AB = B$.

COMMON MISTAKE

In a minterm, a 0 in the row means the variable is complemented. Row $A=0, B=1$ is $A'B$, not AB'.

Studio — $Y = (A + B)' \cdot C$

NOR of A, B feeding AND with C. $Y = 1$ only when $A = B = 0$ and $C = 1$. One minterm: $A'B'C$.

A	B	C	A+B	(A+B)′	Y
0	0	0	0	1	0
0	0	1	0	1	1
0	1	0	1	0	0
0	1	1	1	0	0
1	0	0	1	0	0
1	0	1	1	0	0
1	1	0	1	0	0
1	1	1	1	0	0

EXIT TICKET

1. How many rows for three inputs? 2. Write the minterm for row $A=0, B=1, C=1$. 3. For $Y = A \cdot B + C$, which gate is drawn first?

HOMEWORK

Truth table of $Y = A' \cdot B + C$ plus SOP of the 1-rows. This is a 6-mark board item — practise it once tonight.

L05

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Karnaugh maps — grouping the ones

TODAY YOU WILL BE ABLE TO

- Place minterms on a 2- and 3-variable K-map in Gray-code order.
- Group 2^n ones, including wrap-around.
- Read the simplified product from a group.

LAST LECTURE

SOP writes one product per 1-row. A K-map is that same census drawn so neighbours differ by one bit and cancel.

Adjacent cells — including wrap-around — differ by one bit. That order is **Gray code**. Group 1, 2, 4, 8 ones (powers of two). The variable that changes inside the group disappears.

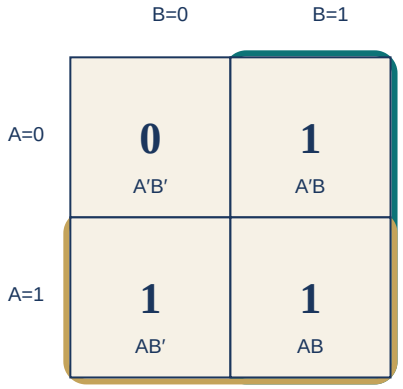


Figure L5.1 — Two-variable map. Right column is B; bottom row is A; so $Y = A + B$.

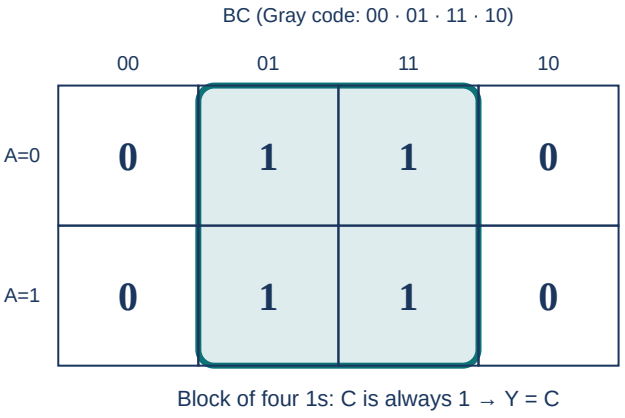


Figure L5.2 — Three-variable map. Columns are $BC = 00, 01, 11, 10$ — never 00, 01, 10, 11.

DO NOW

Simplify $Y = \sum m(0, 1, 2, 3, 5, 7)$. Two groups of four give $Y = A' + C$.

Wrap-around trap: $Y = \sum m(0, 2, 8, 10)$ occupies the four corners. Shared literals are B' and C'; A cancels. Result $Y = B'C'$. Forget wrap-around and you leave four lonely minterms — the classic exam miss.

EXIT TICKET

1. Why must columns be Gray-coded? 2. Legal group sizes? 3. What happens to a variable that changes inside a group?

HOMEWORK

K-map $Y = \sum m(1, 3, 5, 7)$. It should collapse toward C. Check your column order before you celebrate.

L06

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Software Development Life Cycle

TODAY YOU WILL BE ABLE TO

- Name the six SDLC phases and one deliverable each.
- Recite CAPP maintenance with a one-line school example.
- Say why a life cycle is a cycle, not a dead end.

LAST LECTURE

Logic tells a machine the truth. SDLC tells a team how an idea becomes software that people can live with.

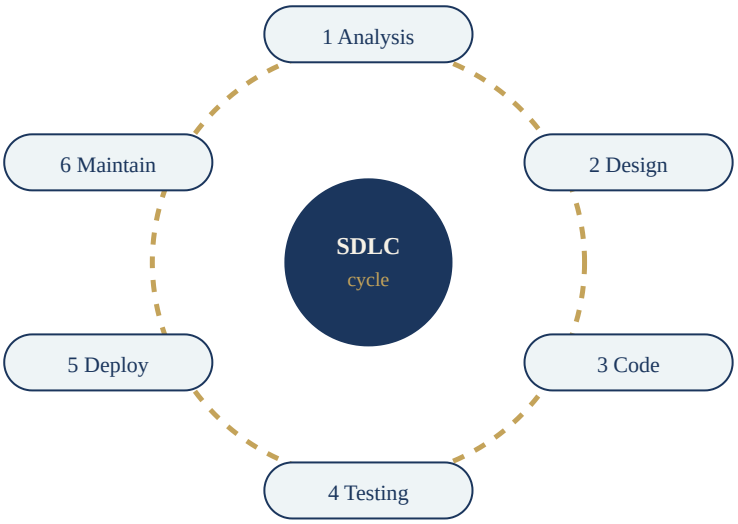


Figure L6.1 — Six phases. Maintenance feeds the next analysis — that is why it is a cycle.

Phase	What you do	What you hand over
1 Analysis	Meet users; list features and constraints	SRS (software requirements)
2 Design	HLD/LLD, screens, ER, UML	Design document
3 Implementation	Code, Git, comments	Working build
4 Testing	Unit, integration, system, UAT	Test report
5 Deployment	Direct, phased, pilot, or parallel	Live software
6 Maintenance	Corrective, adaptive, perfective, preventive	Patches

CAPP maintenance — learn four words, four school pictures:

- **Corrective** — the login button crashes; you fix the bug.
- **Adaptive** — Windows updates; you retarget the installer.
- **Perfective** — students ask for dark mode; you add it.
- **Preventive** — a daily USB backup so a disk death is not a funeral.

FIRST DELIVERABLE

If a one-mark item asks for the first SDLC document, the answer is the **SRS**, not “coding.”

EXIT TICKET

1. First phase and its deliverable? 2. Name the four maintenance types. 3. Pilot versus parallel deployment in one sentence each.

HOMEWORK

Close the book. Write Analyse → Design → Code → Test → Deploy → Maintain and one CAPP example each. Time yourself: under four minutes is ready.

L07

UNIT 1 · COMPUTER SYSTEMS · 45 MIN · INTERMEDIATE / CLASS XI

Waterfall versus Agile — Al-Noor cases

TODAY YOU WILL BE ABLE TO

- Match Waterfall to a frozen, well-specified project.
- Match Agile/Scrum to unclear, evolving features.
- Retell both Al-Noor stories in six sentences.

LAST LECTURE

Six phases, CAPP, SRS. Today: two ways to walk those phases — one way down, or in short sprints.

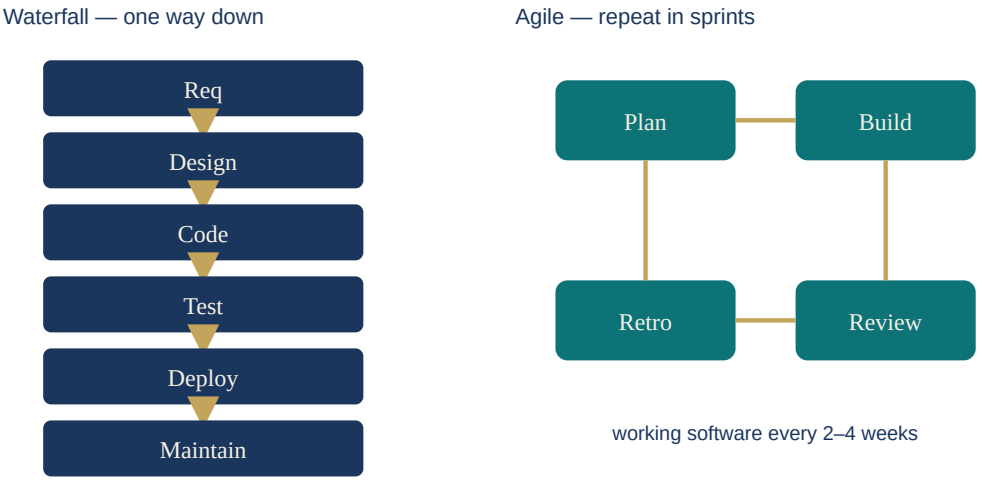


Figure L7.1 — Waterfall finishes a phase before the next begins. Agile repeats Plan → Build → Review → Retro.

Al-Noor Library — why Waterfall worked

1,200 students, 2,400 books, a paper register, lost copies. Requirements were frozen: add/edit books, search, 14-day loans, overdue alerts, printed reports, offline Windows. Team of four, six months, Python + Tkinter + SQLite. Nothing experimental was asked, so Waterfall fitted. Black-box testing found a search crash on misspellings and an overdue alert that ran only if a button was clicked. A two-week **pilot** sat on the librarian's desk. Later: printer-driver fix (corrective), e-book category (adaptive), faster search (perfective), daily USB backup (preventive). Overdue returns fell about 70%.

Al-Noor Student App — why Agile worked

Nobody could list every feature on day one. Five two-week sprints, Flutter, five-person Scrum team. Sprint 1 shipped login + timetable; teachers' names arrived from feedback. Sprint 2 shipped results; parents then asked for full attendance. Sprint 3's SMS gateway died for two days — the team switched to email and kept moving. Sprint 4 connected to the library API in thirty minutes. Sprint 5 shipped homework photos. Five of ten original stories shipped in ten weeks. That flexibility is the point of Agile.

DECISION RULE (4-mark compare)

Pacemaker firmware with a frozen regulator spec → Waterfall (late change can kill). A club-events app that will grow with student requests → Agile. Four lines in a notebook is a complete answer.

EXIT TICKET

1. One project feature that forces Waterfall. 2. What is a sprint? 3. Which Al-Noor project used a pilot, and why?

HOMEWORK

Write a one-page Waterfall vs Agile table from memory: when, risk of late change, documentation, user feedback.

L08

UNIT 1 · COMPUTER SYSTEMS · 45 MIN · INTERMEDIATE / CLASS XI

How machines converse — OSI and TCP/IP

TODAY YOU WILL BE ABLE TO

- Recite the seven OSI layers with the pizza mnemonic.
- Name the encapsulation units: data, segment, packet, frame, bits.
- Map OSI onto the four TCP/IP layers and place TCP, UDP, IP, MAC.

LAST LECTURE

Teams need a life cycle. Distant machines need a conversation plan. That plan is layered.

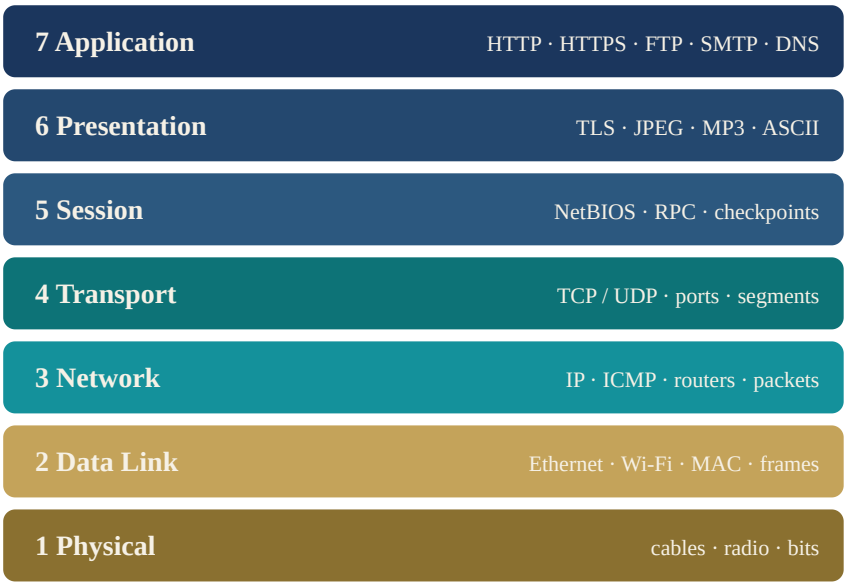


Figure L8.1 — Please Do Not Throw Sausage Pizza Away.

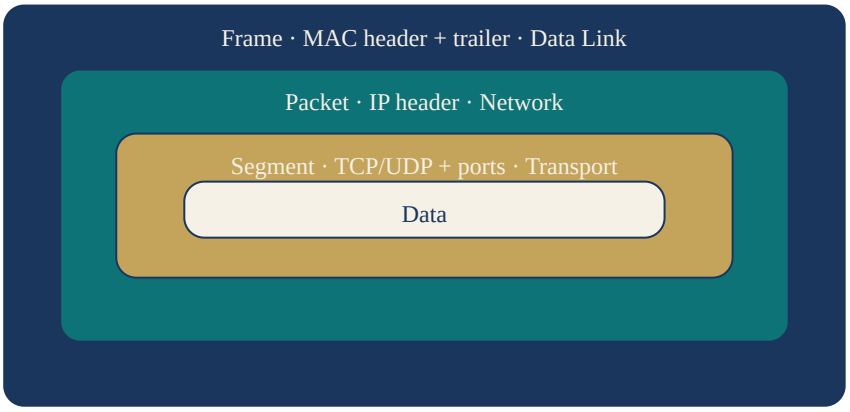


Figure L8.2 — Going down: wrap. Going up: unwrap.

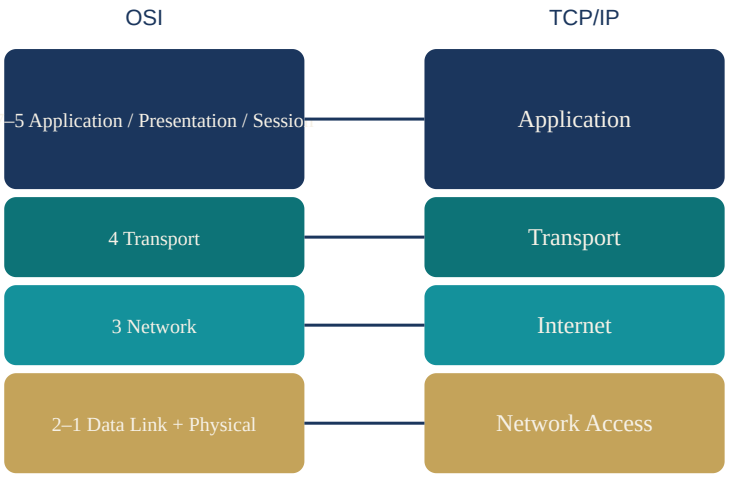


Figure L8.3 — TCP/IP swallows OSI 7–5 as Application and OSI 2–1 as Network Access.

If you see…	Think layer	Why
RJ45, Cat6, voltage, fibre pulse	Physical	Raw medium
MAC, switch, Ethernet frame, CRC	Data Link	One LAN hop
192.168.1.1, ping, router, OSPF	Network	Path across nets
Port 443, handshake, segment	Transport	Which app, reliability
TLS, JPEG, MP3, UTF-8	Presentation	Format / secret / size
Browser, DNS, SMTP	Application	What the user asked

TCP is connection-oriented and reliable (web, email, files). UDP is connectionless and cheap (live video, games, a cricket stream — late frames are useless). An **IP address** is logical; a **MAC address** is burned into the NIC. Routers live at Layer 3; switches at Layer 2; hubs at Layer 1. `ping` uses ICMP at the Network layer.

> **EXIT TICKET**
> 1. Which layer adds ports / IP / MAC? 2. Why might a cricket stream use UDP? 3. Four TCP/IP names?

> **HOMEWORK**
> Explain HTTPS encapsulation in five sentences: data → segment → packet → frame → bits. Then write one protocol for each TCP/IP layer.

L09

UNIT 2 · COMPUTATIONAL THINKING · 45 MIN · INTERMEDIATE / CLASS XI

Think so a machine can follow you

TODAY YOU WILL BE ABLE TO

- Define an algorithm and contrast it with pseudocode.
- Name decomposition, pattern recognition, and abstraction on a school task.
- Judge an algorithm on correctness, efficiency, and clarity.

BLOCK B OPENS

Unit 1 taught machines truth and conversation. Unit 2 teaches you to write steps so clear that a classmate — or Python — can follow them.

An **algorithm** is a finite, language-independent sequence of logical steps. It is the recipe, not the cooked dish. **Pseudocode** writes those steps with IF / FOR / WHILE but no compiler punctuation.

	Algorithm	Pseudocode
Audience	Anyone who can follow logic	Someone about to code
Form	Numbered everyday steps	Structured keywords
Syntax	None required	Conventions, still not executable
Purpose	Get the logic right	Bridge logic to a language

- **Decomposition** — split “learn Spanish” into alphabet, vocabulary, grammar, practice. A school app becomes login, timetable, results.
- **Pattern recognition** — a triangle of stars that grows by one star per row is a loop $i = 1 \ldots n$.
- **Abstraction** — making tea is boil, add, steep, pour. Ignore mug colour and kettle brand.

Move	Strategy
Split “build the app” into login, marks, notices	Decomposition
Every notice has a title, date, body — same shape	Pattern recognition
Ignore button colour when specifying “post a notice”	Abstraction
Write numbered steps before opening VS Code	Algorithm

BOARD TRAP

Ignoring the kettle brand is **abstraction**, not decomposition. Decomposition lists the subtasks. Pattern recognition notices that every hot drink shares “heat → infuse → pour.”

Judge every algorithm on **correctness** (right for every valid input, and it finishes), **efficiency** (time and memory as n grows), and **clarity** (a classmate can maintain it).

EXIT TICKET

1. Define an algorithm in one sentence. 2. Which CT strategy ignores mug colour? 3. Name correctness, efficiency, clarity.

HOMEWORK

Write an algorithm (not Python) for borrowing a library book. Mark one step as abstraction and one as decomposition.

L10

UNIT 2 · ALGORITHMS · 45 MIN · INTERMEDIATE / CLASS XI

Bubble Sort — neighbours swap

TODAY YOU WILL BE ABLE TO

- Trace Bubble Sort on a 5-item list, pass by pass.
- Write the nested-loop pseudocode.
- State one strength and one weakness.

LAST LECTURE

An algorithm is finite steps. Today the steps compare neighbours until the list is ordered.

Walk along the list. Whenever two neighbours are in the wrong order, swap them. After pass k, the k largest items sit at the right. Nested loops give time that grows like n^2.



Figure L10.1 — Trace of {8, 4, 1, 9, 3}. Teal cells are already parked.

```
FOR pass ← 0 TO n−2
    FOR i ← 0 TO n−2−pass
        IF DATA[i] > DATA[i+1] THEN swap
```

Diary of {8, 4, 1, 9, 3}: pass 1 ends {4, 1, 8, 3, **9**}; pass 2 ends {1, 4, 3, **8, 9**}; pass 3 ends {1, 3, **4, 8, 9**}; pass 4 confirms.

Strengths: easy to see; in-place; stable (equal keys keep their order). **Weaknesses:** many swaps; hopeless for large n.

Studio list $\{6, 2, 5, 1\}$: pass 1 → $\{2, 5, 1, 6\}$ (three swaps); pass 2 → $\{2, 1, 5, 6\}$; pass 3 → $\{1, 2, 5, 6\}$.

EXIT TICKET

1. After pass 1, where does the largest item sit? 2. Why is the inner loop shorter each pass? 3. Is Bubble Sort stable?

HOMEWORK

Trace {8, 3, 6, 2} on paper. Show the list at the end of each pass. Count swaps.

L11

UNIT 2 · ALGORITHMS · 45 MIN · INTERMEDIATE / CLASS XI

Selection Sort — park the minimum

TODAY YOU WILL BE ABLE TO

- Trace Selection Sort on the same list as Lecture 10.
- Explain why it usually swaps less than Bubble Sort.
- State that it is not stable.

LAST LECTURE

Bubble Sort swaps neighbours and parks the largest at the right. Selection Sort parks the next minimum at the left — at most one write per pass.

On pass k, find the minimum of the unsorted tail and swap it into index k. Still $\Theta(n^2)$ comparisons, fewer writes, **not stable**.

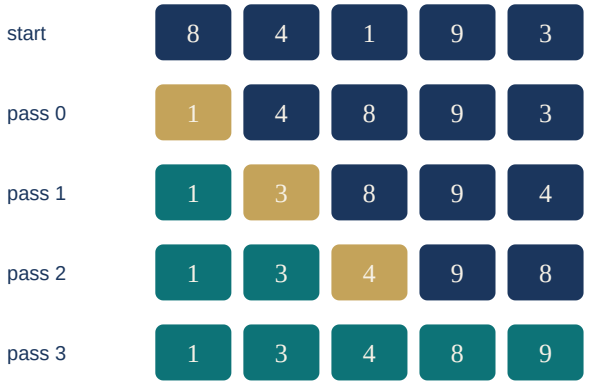


Figure L11.1 — Same list {8, 4, 1, 9, 3}. Gold is the newly parked minimum.

- Pass 0: smallest of the whole list is 1; swap with 8 → {1, 4, 8, 9, 3}.
- Pass 1: smallest of the tail is 3; swap with 4 → {1, 3, 8, 9, 4}.
- Pass 2: smallest of the tail is 4; swap with 8 → {1, 3, 4, 9, 8}.
- Pass 3: smallest of the tail is 8; swap with 9 → {1, 3, 4, 8, 9}.

On $\{6, 2, 5, 1\}$ Selection finds min 1 and swaps once with $6 \rightarrow \{1, 2, 5, 6\}$. Watch equal keys: sort {9, 4, 7, 4, 1}. Bubble Sort should keep the first 4 ahead of the second. Selection Sort may swap their order — that single observation is a 3-mark “disadvantage” answer.

TEXTBOOK COUSIN

Some traces swap as soon as a smaller value is seen. Standard selection finds the *index* of the minimum, then swaps once. In the paper, follow the version your teacher traced.

EXIT TICKET

1. How many swaps per pass in standard Selection Sort? 2. Why is it not stable? 3. Same time class as Bubble Sort?

HOMEWORK

Sort {9, 4, 7, 4, 1} with both algorithms. Circle the two 4s and see who keeps their order.

L12

UNIT 2 · ALGORITHMS · 45 MIN · INTERMEDIATE / CLASS XI

Linear search and binary search

TODAY YOU WILL BE ABLE TO

- Trace linear search on unsorted data.
- Compute mid and discard a half on a sorted list.
- Say when binary search is simply wrong.

LAST LECTURE

Two sorts, both n^2. Searching asks a different question: is this value here, and where?

Linear search checks items $1, 2, \ldots, n$. Works on unsorted data. Worst case n probes.

Binary search requires a sorted array. Each step uses

$$\text{mid} = \left\lfloor \frac{\text{low} + \text{high}}{2} \right\rfloor.$$

If the target is smaller, set $\text{high} = \text{mid} - 1$; if larger, $\text{low} = \text{mid} + 1$. About $\log_2 n$ probes — twenty looks for a million items.

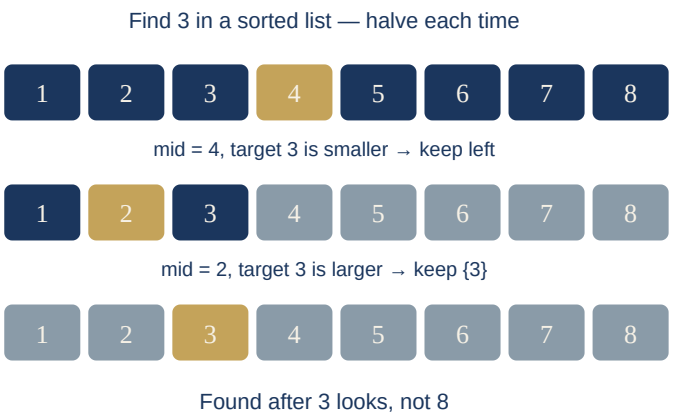


Figure L12.1 — Find 3. Gold is mid; grey cells have been discarded.

Find 11 in $\{2, 5, 7, 11, 13, 17, 19\}$: $\text{mid} = \lfloor (0 + 6)/2 \rfloor = 3$, hit on the first probe. Find 2 (left edge): three looks. Find 12 (absent): finish with $\text{low} > \text{high}$ and report not found — write that failure case so you do not invent a crash.

	Linear	Binary
Needs sorted data?	No	Yes
Idea	One-by-one	Halve the interval
Best for	Small or unsorted; one-off search	Large static sorted lists
Growth	$O(n)$	$O(\log n)$

COMMON MISTAKE

Binary search on an unsorted list is not “a bit worse.” It is wrong: the discarded half may hold the target. {9, 1, 4, 7} is a 2-mark trap.

EXIT TICKET

1. One limitation of binary search. 2. Write the mid formula. 3. Why does linear search still matter?

HOMEWORK

Binary-search 6 in {1,2,3,4,5,6,7,8,9}. Write every low, high, mid. Then state what happens if the list is shuffled.

L13

UNIT 3 · PROGRAMMING · 45 MIN · INTERMEDIATE / CLASS XI

Python, the IDE, and input() is a string

TODAY YOU WILL BE ABLE TO

- Say why Python is a high-level language.
- Run a three-line program in VS Code.
- Convert input() with int() or float() before arithmetic.

BLOCK C OPENS

A computer does exactly what you said, including the mistakes. Python is this course's high-level language because it reads almost like English.

Low-level sits next to the machine (binary, assembly). **Mid-level** (C) lets you touch memory. **High-level** (Python, Java) hides the hardware so you think about the problem. C++ is for speed; Java's promise is "write once, run anywhere"; Python's promise is readable syntax and a huge library world.

An **IDE** is one workplace: editor, colour, run, debugger. VS Code becomes an IDE when you install the Microsoft Python extension. Save as `program.py`. Indentation is syntax: **4 spaces** per block. Mix tabs and spaces and you will meet `IndentationError`.

```
# a comment — the interpreter ignores it
print("Hello, Intermediate")
print(100)
```

Execution is top to bottom. That default order is **sequence**.

Type	Keyword	Example
Integer	int	20, −3
Float	float	5.7
String	str	"Ali", '2026'
Boolean	bool	True
List / tuple / dict	list, tuple, dict	[2,4], (10,), {"name": "Ahmed"}

`input()` always returns a **string**. Convert: `age = int(input("Age: "))`. Forget the conversion and `"10" + "20"` quietly gives `"1020"`.

THE CLASSIC FIRST-YEAR CRASH

Using input() in arithmetic without int() or float(). Names: letters, digits, underscore; start with a letter or _; never a keyword. `my-var` is illegal (the hyphen is minus).

EXIT TICKET

1. What does input() return? 2. Standard indent? 3. Why is my-var illegal?

HOMEWORK

Type a program that asks for two numbers, converts them with float(), and prints the sum. Break it on purpose by removing float() and read the error.

L14

UNIT 3 · PROGRAMMING · 45 MIN · INTERMEDIATE / CLASS XI

Operators — including the bitwise family

TODAY YOU WILL BE ABLE TO

- Use //, %, and ** correctly.
- Contrast = (store) with == (test).
- Compute AND, OR, XOR, and a left shift on small integers.

LAST LECTURE

input() is a string; 4 spaces; sequence. Today the symbols that sit between values.

Family	Symbols	Notes
Arithmetic	+ − * / // % **	$9//2 = 4$, $9\%2 = 1$, $2^3 = 8$
Assignment	= += −= *=	After a = 10; a += 5, a is 15
Relational	== != > < >= <=	Return True / False
Logical	and · or · not	Combine conditions
Membership	in · not in	"P" in "Python" is True
Bitwise	& | ^ ~ << >>	Same truth tables as Unit 1 gates

Bitwise worked example. $a = 10 = 1010_2$, $b = 6 = 0110_2$:

$$a \wedge b = 0010_2 = 2, \quad a \vee b = 1110_2 = 14, \quad a \oplus b = 1100_2 = 12, \quad a \ll 1 = 20 = a \cdot 2.$$

Also $\sim x = -x - 1$ in two's complement, so $\sim 10 = -11$. If a question shows `0b1010`, the `0b` prefix is notation, not part of the value.

PREDICT (True/False)

$9//2 == 4 \cdot 9\%2 == 1 \cdot$ `"Py" in "Python"` $\cdot$ `not (True and False)`. All four are True.

COMMON MISTAKE

= stores a value. == tests equality. Mixing them inside an if is a logic bug that looks like a typo.

EXIT TICKET

1. What is 9 // 2? 2. Bitwise AND of 10 and 6? 3. Difference between = and ==?

HOMEWORK

Line up 10 (1010) and 6 (0110) on paper. Compute AND, OR, XOR, and 5 << 2 (= 20).

L15

UNIT 3 · PROGRAMMING · 45 MIN · INTERMEDIATE / CLASS XI

Selection — if, elif, else

TODAY YOU WILL BE ABLE TO

- Draw a selection flowchart with exactly one path taken.
- Write if / elif / else for a three-way choice.
- Nest a guard (for example refuse ÷ 0).

LAST LECTURE

Operators decide True/False. Selection uses that answer to pick a path.

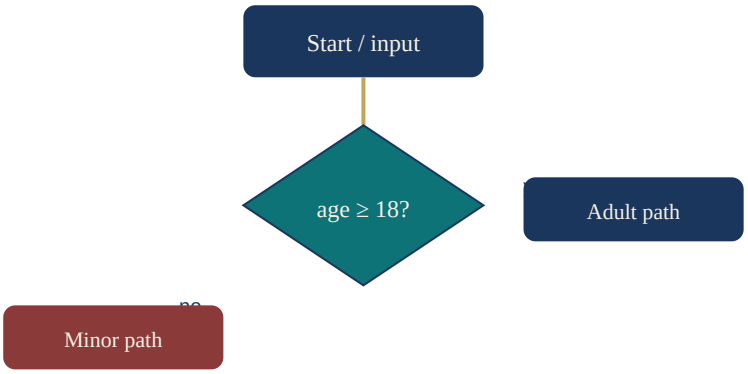


Figure L15.1 — Selection: exactly one path runs.

for waits until Lecture 16. Today: `if`, `elif`, `else`. The colon opens a block; the indent is the block.

```
a = float(input("First number: "))
b = float(input("Second number: "))
op = input("Choose + - * / : ").strip()
if op == "+":
    print(a + b)
elif op == "-":
    print(a - b)
elif op == "*":
    print(a * b)
elif op == "/":
    if b == 0:
        print("Cannot divide by zero.")
    else:
        print(a / b)
else:
    print("Unknown operator.")
```

The nested if inside division is selection inside selection — the unit's nested-if idea in a useful costume.

EXIT TICKET

1. How many paths of an if/elif/else run? 2. Why does / need a nested guard? 3. What does strip() remove?

HOMEWORK

Write even/odd: read an int, print Even if n % 2 == 0 else Odd. Then add a third branch for zero.

L16

UNIT 3 · PROGRAMMING · 45 MIN · INTERMEDIATE / CLASS XI

Repetition — for, while, break, continue

TODAY YOU WILL BE ABLE TO

- Choose for when the trip count is known.
- Choose while until a condition dies.
- Trace a while-loop that never dies.

LAST LECTURE

Selection picks one path. Repetition walks a path many times.

for when the count is known: `range(start, stop, step)` — stop is exclusive. **while** until a condition becomes False. `break` leaves now; `continue` skips this iteration.

```
for i in range(1, 6):
    print(i)          # 1 2 3 4 5

i = 1
while i < 6:
    print(i)
    i += 1            # without this line the loop never dies
```

LOGICAL ERROR, NOT A CRASH

The textbook infinite loop: `i = 1; while i < 6: print(i); i -= 1`. Trace: 1, 0, −1… ***i*** never reaches 6. The program is legal and will run until you kill it. Always ask: “What makes the condition False, and does every path reach it?”

A one-shot guess uses `random.randint` and a single if. A PIN loop (next lecture) uses while + a bool flag.

EXIT TICKET

1. Loop when the count is known? 2. What does continue do? 3. Why does i -= 1 inside while i < 6 never finish?

HOMEWORK

Print a triangle of stars, 1 to 5, with a nested for. Then rewrite it with a while. Same output, two control stories.

L17

UNIT 3 · PROGRAMMING · 45 MIN · INTERMEDIATE / CLASS XI

Libraries, errors, and the ATM studio

TODAY YOU WILL BE ABLE TO

- Import math, random, and datetime.
- Name syntax, runtime, and logical errors.
- Read an ATM skeleton as sequence + selection + repetition.

LAST LECTURE

for / while / flags. Today we borrow other people's code and name the three ways a program can be wrong.

```
import math, random, datetime
print(math.sqrt(25), math.pi)
secret = random.randint(1, 10)
now = datetime.datetime.now().strftime("%d-%m-%Y")
```

Third-party libraries need `pip install` (NumPy, Pandas, Matplotlib). Built-ins are already housemates.

Error	When	Example
Syntax	Will not start	Missing), bad indent, if without :
Runtime	Dies while running	10/0, int("abc")
Logical	Runs and lies	i -= 1 in a climbing loop

Read a traceback from the **bottom**: last line is the disease.

ATM skeleton — Unit 3 in one costume

```
PIN = "1234"
balance = 10000.0
tries = 3
ok = False
while tries > 0 and not ok:
    entered = input("Enter PIN: ")
    if entered == PIN:
        ok = True
    else:
        tries -= 1
        print("Wrong PIN!", tries, "left.")
if not ok:
    print("Card locked.")
else:
    running = True
    while running:
        print("1 Check  2 Deposit  3 Withdraw  4 Exit")
        choice = input("Choose: ")
        if choice == "1":
            print(f"Balance: Rs. {balance:.2f}")
        elif choice == "2":
            amt = float(input("Deposit: "))
            if amt > 0:
                balance += amt
        elif choice == "3":
            amt = float(input("Withdraw: "))
            if 0 < amt <= balance:
                balance -= amt
            else:
                print("Rejected.")
        elif choice == "4":
            running = False
```

Sequence (start values) + selection (menu) + repetition (PIN loop, menu loop) + bool flags. If you can add a transfer or a 20-try lockout, you own the unit.

> **EXIT TICKET**
> 1. Three error types? 2. Where do you start reading a traceback? 3. Which structure is the PIN loop?

> **HOMEWORK**
> Type the ATM. Add deposit and a rejected-overdraw message. Then break the PIN on purpose and read the traceback from the bottom.

L18

UNIT 4 · DATA AND ANALYSIS · 45 MIN · INTERMEDIATE / CLASS XI

Data, DBMS, and the four keys

TODAY YOU WILL BE ABLE TO

- Turn raw marks into one sentence of information.
- Define database, DBMS, and RDBMS.
- Place candidate, primary, alternate, and foreign keys.

BLOCK D OPENS

Eighty, seventy-five, ninety is data. “The class average is 80%” is information. A database is the organised cupboard that keeps those facts consistent.

	Data	Information
Definition	Raw facts	Processed, meaningful facts
Example	85, 90, 75	Average = 83.3%
Use	Little, alone	Supports a decision

A **DBMS** (Access, MySQL, Oracle, SQL Server) inserts, updates, deletes, and retrieves with less redundancy, better consistency, security, and integrity. An **RDBMS** stores related tables and is queried with SQL.

A **table** is rows × columns. A **record** is a row. A **field** is a column.

Key	Role
Candidate	Any unique non-null identifier — a table may have several
Primary	The chosen candidate; exactly one; never NULL
Alternate	A leftover candidate (still unique)
Foreign	Points at another table’s primary key; may repeat

ONE-LINER

Primary key: unique, not null, one per table. Foreign key: lives in the child, points home.

EXIT TICKET

1. DBMS besides Access? 2. Define a foreign key. 3. Can a table have two primary keys?

HOMEWORK

For a Student table list two candidate keys and choose a primary key. Write one sentence why CNIC is a poor classroom PK (privacy).

L19

UNIT 4 · DATA AND ANALYSIS · 45 MIN · INTERMEDIATE / CLASS XI

ER models and the three cardinalities

TODAY YOU WILL BE ABLE TO

- Draw entity rectangles and name attributes.
- Recognise 1:1, 1:M, and M:N.
- Resolve M:N with a junction table.

LAST LECTURE

Keys identify rows. Relationships say how rows of two tables belong together.

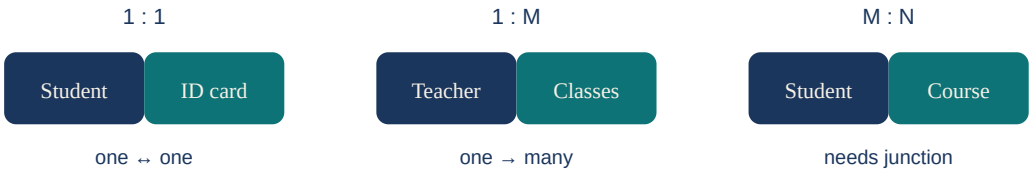


Figure L19.1 — Only M:N needs a junction table.

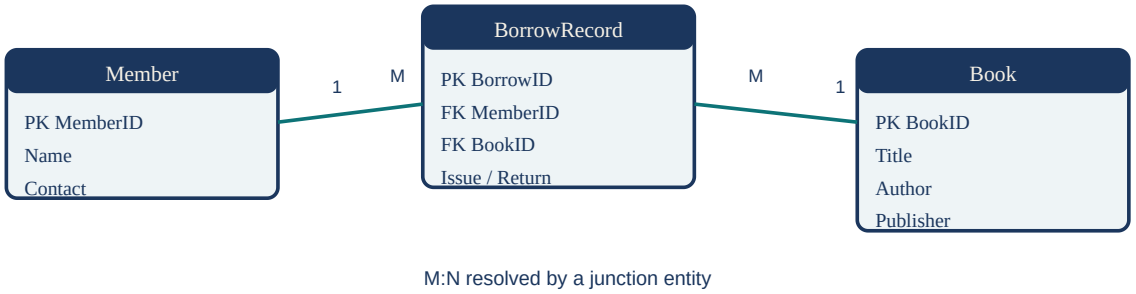


Figure L19.2 — Library ER: Member and Book are M:N, resolved as BorrowRecord.

A student has one ID card (1:1). A teacher teaches many classes (1:M). Students enrol on many courses and a course has many students (M:N) — that last pair needs a junction (Enrolment) holding both foreign keys plus extras such as date.

Clinic in 15 minutes. Patient and Doctor are M:N, resolved as Appointment(AppointmentID PK, PatientID FK, DoctorID FK, VisitDate, Fee). Customer 1—M Order needs *no* junction: one order belongs to one customer. Students over-use junction tables.

EXIT TICKET

1. ER rectangle is which concept? 2. Which relationship needs a junction? 3. Name the library junction entity.

HOMEWORK

On paper, Bus / Route / Student. A bus serves one route; many students ride many buses across a week — that last link may need a Ride junction with a date.

L20

UNIT 4 · DATA AND ANALYSIS · 45 MIN · INTERMEDIATE / CLASS XI

Referential integrity and Access objects

TODAY YOU WILL BE ABLE TO

- State referential integrity with a school example.
- Warn about cascade delete.
- Name table, form, query, report.

LAST LECTURE

M:N is two 1:M links through a junction. Those links only work if every foreign key points at a living parent.

Referential integrity: every foreign key must match a living parent. You cannot store marks for StudentID 999 if that student does not exist.

Cascade update rewrites child FKs when the parent PK changes. *Cascade delete* removes child rows when the parent is deleted — use it like a chainsaw. Cascade-delete of a doctor wipes appointment history; better to forbid the delete or mark the doctor inactive.

Access objects: **table** stores, **form** enters, **query** asks, **report** prints. Create a table in Datasheet or Design view; pick datatypes (Short Text, Number, Date/Time, Currency, Yes/No, AutoNumber for a surrogate PK).

EXIT TICKET

1. Define referential integrity. 2. Cascade update in one sentence. 3. Which object is for friendly entry?

HOMEWORK

Write a 3-mark caution: why cascade-delete on Doctor is dangerous in the clinic schema.

L21

UNIT 4 · DATA AND ANALYSIS · 45 MIN · INTERMEDIATE / CLASS XI

Queries, totals, and charts

TODAY YOU WILL BE ABLE TO

- Put AND on the same criteria row and OR on different rows.
- Use Group By with Count or Avg.
- Pick line, pie, or bar for the story.

LAST LECTURE

Tables store; forms enter; queries ask. Today the Design grid — the usual practical-exam discriminator.

Need	Design grid
CS students only	Department Criteria: "Computer Science"
IT or CS	Criteria / or: two department values
Aslam in IT	Name and Department on the SAME row
Headcount per dept	Totals on; Department Group By; StudentID Count
Average marks per dept	Department Group By; Marks Avg

AND is horizontal (one row). OR is vertical (criteria / or). Totals Σ: Group By + Count / Avg / Sum / Min / Max.

Line charts show trends (monthly attendance). Pies show parts of a whole (pass/fail). Bars compare named categories (marks by department).

RECITE

AND same row. OR different rows. Chart for a monthly trend = line.

EXIT TICKET

1. Aggregate for a mean? 2. Chart for a monthly trend? 3. How do you write OR in Query Design?

HOMEWORK

On paper, design a grouped query: average fee per doctor from Appointment. Name the Group By field and the aggregate.

L22

UNIT 5 · APPLICATIONS · 45 MIN · INTERMEDIATE / CLASS XI

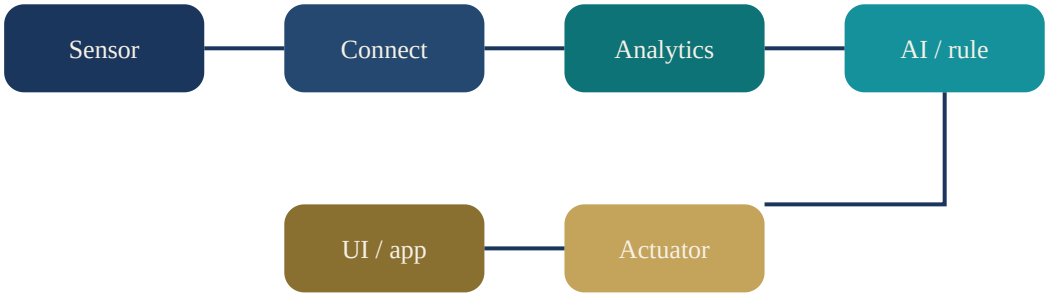
IoT collects, analytics explains, AI decides

TODAY YOU WILL BE ABLE TO

- Keep IoT, analytics, and AI as a loop, not as synonyms.
- Name sensor, connectivity, processing, actuator, cloud, UI.
- Walk a Sindh irrigation chain in six labelled sentences.

BLOCK E OPENS

Computing has left the lab. Today three cooperating technologies — and the exam trap of treating the three words as the same thing.



Collect → carry → explain → decide → act → show

Figure L22.1 — Smart irrigation: moisture sensor → radio → cloud analysis → valve → farmer's phone.



Raw marks become a pass/fail chart only after cleaning

Figure L22.2 — Raw records become a decision only after cleaning.

	IoT	Analytics	AI
Job	Sense and actuate	Find patterns	Decide or predict
Pieces	Sensors, actuators, links, cloud, UI	Collect–store–clean–analyse–visualise	Models, learning
Example	Soil probe, smartwatch	Pass/fail pie chart	Face unlock, recommendation

Sensor ≠ actuator. A moisture probe does not open the tap; the valve does. Connectivity (Wi-Fi, Bluetooth, 4G/5G, LoRa) carries the message. Cloud stores the history. The UI is how a human looks and commands.

Memorise the smart-home skeleton: sensor → Wi-Fi → hub/cloud → analytics vs set-point → rule/AI → actuator → phone UI. If Wi-Fi dies the chain breaks — that one failure sentence shows you understand dependency.

LONG-ANSWER SKELETON (6 marks)

A farmer in Sindh wants automatic irrigation. One mark each: sensor, link, decide/process, valve, farmer UI, and where analytics/AI sit on the decision side.

EXIT TICKET

1. Watch → phone heart-rate is mainly which tech? 2. Device that judges dry soil vs opens the valve? 3. Duplicate rows are which analytics step?

HOMEWORK

Write the irrigation paragraph in six labelled sentences without looking. Then add the failure sentence (Wi-Fi dies).

L23

UNIT 5 · APPLICATIONS · 45 MIN · INTERMEDIATE / CLASS XI

Sources, impacts, and assistive technology

TODAY YOU WILL BE ABLE TO

- Classify primary, secondary, and tertiary sources.
- Write a 4-mark impact paragraph: help + harm + responsible action.
- Name four assistive-tech families.

LAST LECTURE

IoT collects, analytics speaks, AI acts. Today: can you trust the story, and who does computing include?

Kind	Meaning	Example
Primary	First-hand original	Interview, NADRA CNIC capture
Secondary	Already explained	Textbook, PBS census summary
Tertiary	Quick digest	Encyclopedia, many Wikipedia leads

Exhibit	Verdict
Your interview with the librarian	Primary source / primary data
PBS graph of literacy rates	Secondary
Wikipedia lead paragraph	Tertiary; start only
Anonymous “forward this” image	Unreliable — no author or date
Government .gov.pk PDF, 2024	Usually reliable secondary

Impact formula: **field + one concrete help + one concrete harm + one responsible action**. “In education, recorded lessons let a student replay a proof (help), but six-hour screen days blunt attention (harm). A responsible student uses the recording for the hard ten minutes, then walks.”

Assistive tech — mobility (wheelchair), sensory (NVDA screen reader), communication (picture boards), cognitive (audiobooks). They are how computing keeps its promise to everyone. Importance: education access, communication, independence, work, social life.

EXIT TICKET

1. Interview is which source? 2. Two reasons a website is unreliable? 3. What does a screen reader do?

HOMEWORK

Classify five sources actually on your phone. Then write one impact paragraph for healthcare using the four-sentence formula.

L24

UNIT 6 · DIGITAL LITERACY · 45 MIN · INTERMEDIATE / CLASS XI

Digital literacy — find, judge, collect, show

TODAY YOU WILL BE ABLE TO

- Use quotes, filetype:, and site: in a search.
- Tell qualitative from quantitative, primary data from secondary data.
- Design an ethical 90-minute mini inquiry.

LAST LECTURE

Sources can lie. Literacy is the filter: find, judge, collect, and show — without stealing words or leaking private data.



Figure L24.1 — From a sharp question to a final artefact.

	Qualitative	Quantitative
Nature	Words, reasons, feelings	Numbers you can count
Ask	Why / how	How many / how much
Example	“Notifications break my focus”	4 hours/day, 72% distracted

Primary data is what you gather for this question. **Secondary data** already exists (a PBS table). A NADRA record is a primary *source*; a published NADRA statistic you download is secondary *data for you*.

Advanced search: quotes for a phrase, `filetype:pdf` for reports, `site:.edu` or `site:.gov.pk` for a domain family. Best PDF query for internet use in Pakistan: `"internet usage in Pakistan" filetype:pdf`.

Ninety-minute inquiry: one sharp question · two advanced searches · eight mixed items (no CNIC, no phone) · 15 classmates with permission · one chart · one recommendation aimed at a real audience (principal, parents, or students).

NOT AN ARTEFACT

A decorated Canva file is not an artefact if it never answers the question. Evidence → claim → recommendation. Copying without credit is plagiarism. Asking permission is ethics, not politeness.

EXIT TICKET

1. Digital literacy in one sentence. 2. What does site: do? 3. Your own survey replies — primary or secondary data?

HOMEWORK

Write eight survey items on weekend internet hours. Mark which are qualitative. Remove any personal-id field. State your sample and your permission sentence.

L25

ALL UNITS · CLOSED BOOK · 60–70 MIN · INTERMEDIATE / CLASS XI

Revision sitting — cards, burglar alarm, 40-mark paper

TODAY YOU WILL BE ABLE TO

- Recite the seven cards without notes.
- Build Y = A · (D + W) and poke eight rows.
- Sit the 40-mark rehearsal and mark it the same evening.

COURSE CLOSES

Twenty-four sittings. Today you produce traces, not highlights.

Burglar-alarm studio

Armed switch A, door D, window W. Siren

$$Y = A \cdot (D + W).$$

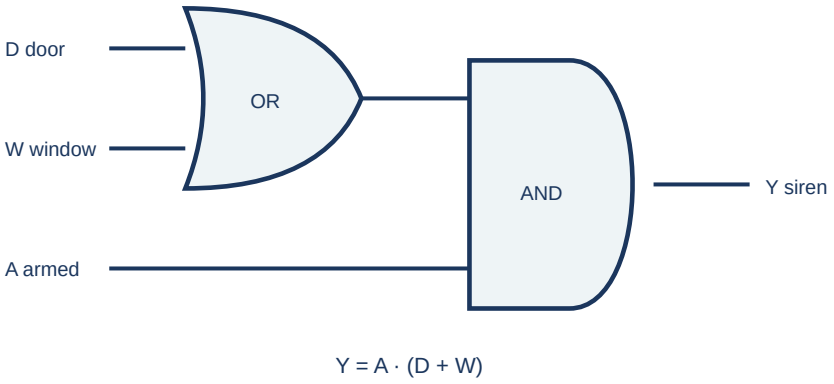


Figure L25.1 — OR of the contacts, then AND with the armed switch.

If $A = 0$ the house stays quiet. Build it in LogiSim and poke all eight rows.

Recitation cards

A Gates. AND all 1s · OR any 1 · NOT flip · NAND not-all-1s · NOR only all 0s · XOR different · XNOR same · rows 2^n.

B SDLC. Analyse → Design → Code → Test → Deploy → Maintain (CAPP).

C OSI. Please Do Not Throw Sausage Pizza Away. ping = ICMP = layer 3.

D Algorithms. Bubble neighbours · Selection parks min · Linear any list · Binary sorted only · $\text{mid} = \lfloor (lo + hi)/2 \rfloor$.

E Python. input → str · indent 4 · = vs == · syntax / runtime / logic.

F Data. PK unique · FK points home · M:N + junction · AND same row · OR different rows.

G Society. Collect / explain / decide · sensor ≠ actuator · consent · no plagiarism.

40-mark rehearsal (60 minutes, closed book)

A (10×1). Digital levels? XOR 1 when? First SDLC phase? TCP/IP layer count? Search that needs sorted data? `input()` type? ER rectangle? Soil dryness device? Your own survey replies? `site:` does what?

B (6×2). Analog vs digital · when Waterfall · why binary search fails on {9, 1, 4, 7} · = vs == · referential integrity · NADRA interview vs textbook paragraph.

C (3×6). Truth table of $Y = A' \cdot B + C$ plus SOP · Bubble-sort trace of {6, 2, 5, 1} · Sindh irrigation chain naming sensor, link, analytics, AI, actuator, UI.

Scheme A: two values; different; analysis; 4; binary; str; entity; sensor; primary; restrict domain.

Scheme B–C (short). Analog continuous / digital two levels. Waterfall when requirements are frozen. Binary search needs order. = stores, == tests. FK must match a living PK. Interview = primary; textbook = secondary. C1 eight rows + SOP. C2 passes → {2,5,1,6}, {2,1,5,6}, {1,2,5,6}; Selection swaps less. C3 six labelled sentences.

Three days without a teacher

Day 1. Seven 2-input tables from a blank page. K-map $Y = \sum m(1, 3, 5, 7)$. OSI both ways. HTTPS encapsulation in five sentences.

Day 2. Both sorts on {8, 3, 6, 2}. Binary-search 6 in {1..9}. Calculator + guess game; break them; read the traceback.

Day 3. Library or clinic ER. AND vs OR query rows. Irrigation long answer. Classify five phone sources. Sit this paper again. Target 32 / 40.

> **READY MEANS**
> A ready Intermediate student scores 28+ and can explain every miss. The board paper rewards people who can do a truth table, trace a sort, write a small program, draw an ER fragment, and judge a source.

If you have worked the amber boxes twice, you are the teacher you were waiting for. Take Lecture 25 again in a month.